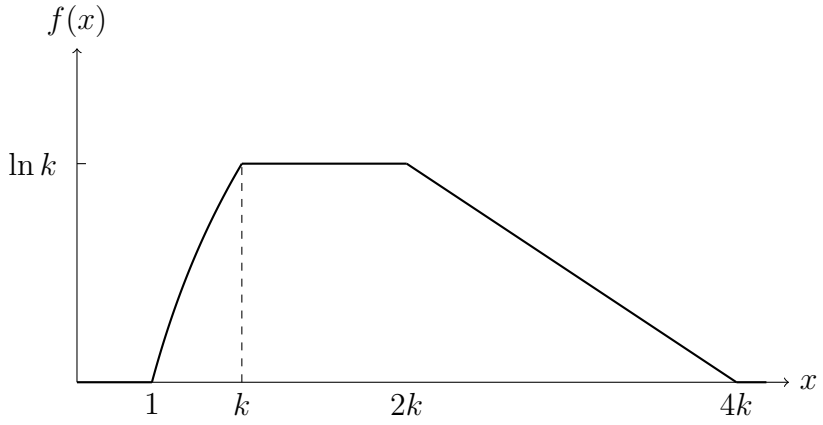
(i)



(ii)

$$\begin{cases} a - 2bk = \ln k \\ a - 4bk = 0 \end{cases} \implies \begin{cases} a = 2\ln k \\ b = \frac{\ln k}{2k} \end{cases}$$

Area:

$$\int_1^k \ln x \, dx + \int_k^{2k} \ln k \, dx + \int_{2k}^{4k} (a - bx) \, dx = 1$$

$$(k \ln k - k + 1) + k \ln k + k \ln k = 1$$

$$\implies k = e^{\frac{1}{3}}$$

$$a = \frac{2}{3}, \; b = \frac{1}{6\, e^{\frac{1}{3}}}$$

(iii)

$$e^{\frac{1}{3}} > 1$$

$$\int_1^k f(x) \, dx = k \ln k - k + 1 = 1 - \frac{2k}{3} < \frac{1}{2}$$

$$\int_1^{2k} f(x) \, dx = 1 - \frac{2k}{3} + \frac{k}{3} = 1 - \frac{k}{3} > \frac{1}{2}$$

Median m lies in $[k, 2k]$

$$\int_1^m f(x) \, dx = \frac{1}{2}$$

$$\int_1^k \ln x \, dx + \int_k^m \ln k \, dx = \frac{1}{2}$$

$$1 - \frac{2k}{3} + (m-k)\ln k = \frac{1}{2}$$

$$m = 3e^{\frac{1}{3}} - \frac{3}{2}$$

This solution was submitted by **Jon**

Published and refined by **Jon**

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